

GLADSTONE AP, Australia

WMO: 943810

Lat: 23.8697S

Lon: 151.2214E

Elev: 55

StdP: 14.67

Time Zone: 10.00 (E10)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	48.1	50.5	25.9	19.9	64.8	30.6	24.8	62.4	19.0	71.4	17.8	71.2	6.0	210	0.382

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	11.4	89.8	75.0	87.8	74.5	86.4	73.8	78.8	84.8	77.8	83.8	76.9	83.0	12.6	40

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
77.2	141.7	81.9	76.1	136.5	81.4	75.0	131.6	80.9	42.4	85.1	41.4	83.9	40.4	83.0	86.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
22.6	19.9	17.7	43.1	95.6	1.5	2.6	42.0	97.4	41.2	99.0	40.4	100.4	39.3	102.3
			37.9	81.0	1.7	1.8	36.6	82.3	35.6	83.3	34.6	84.3	33.4	85.6
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	72.3	79.8	79.6	78.0	74.0	68.7	64.8	63.4	64.4	68.8	72.6	75.9	78.6	(6)
	DBStd	6.73	2.58	2.47	2.44	2.63	3.62	3.98	3.64	3.64	3.39	3.26	2.98	3.02	(7)
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0	(8)
	HDD65	196	0	0	0	0	11	51	74	54	6	0	0	0	(9)
	CDD50	8155	924	829	867	721	578	445	415	445	563	702	778	888	(10)
	CDD65	2876	459	409	402	271	124	46	24	34	119	237	328	423	(11)
	CDH74	21266	4401	3850	3273	1543	446	80	47	109	494	1143	2160	3720	(12)
	CDH80	4921	1258	1125	784	185	16	1	1	5	53	153	372	968	(13)
(14) Wind	WSAvg	8.7	10.8	10.1	9.5	8.6	7.4	6.9	6.9	7.2	7.8	9.3	10.0	10.2	(14)
(15) Precipitation	PrecAvg	36.6	5.8	6.1	4.0	1.9	2.2	1.5	1.4	1.3	0.9	2.2	2.8	5.2	(15)
	PrecMax	68.2	25.2	28.0	14.6	9.8	12.4	8.7	6.7	5.2	5.1	10.9	8.6	20.0	(16)
	PrecMin	17.0	0.2	0.3	0.5	0.0	0.0	0.0	0.0	0.0	0.0	0.1	0.0	0.1	(17)
	PrecStd	11.4	5.1	6.4	3.7	2.0	2.6	1.8	1.7	1.2	1.1	2.2	2.0	4.5	(18)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	92.1	91.7	89.7	85.9	81.8	78.5	77.8	80.3	85.2	87.2	89.7	92.6	(19)
		MCWB	76.8	76.1	74.4	71.0	67.8	64.9	61.8	63.2	68.3	70.1	69.9	75.2	(20)
	2%	DB	89.0	88.7	86.9	83.5	79.7	76.1	75.1	76.7	81.0	83.7	85.9	88.8	(21)
		MCWB	76.3	75.6	74.1	69.9	66.4	64.1	61.9	62.7	65.4	70.1	71.7	74.4	(22)
	5%	DB	86.9	87.1	85.4	82.0	78.2	74.5	73.4	74.7	78.6	81.4	83.7	86.4	(23)
		MCWB	74.9	74.8	73.2	69.3	65.4	63.1	61.2	61.8	64.9	69.4	70.9	73.8	(24)
	10%	DB	85.4	85.7	84.1	80.8	76.6	73.0	71.8	73.1	76.7	79.5	81.9	84.7	(25)
		MCWB	74.0	74.1	72.4	68.7	64.6	62.3	60.6	60.9	64.2	68.1	70.0	72.9	(26)
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	80.2	79.4	79.1	76.6	72.4	70.2	69.4	69.1	72.2	74.7	77.0	80.1	(27)
		MCDB	86.6	86.0	84.4	81.2	77.9	73.7	72.4	74.8	78.9	81.1	83.2	84.9	(28)
	2%	WB	78.8	78.2	77.3	74.2	70.2	68.1	66.6	67.2	70.1	73.2	75.1	78.1	(29)
		MCDB	84.9	84.3	83.1	79.3	75.9	72.2	71.0	72.6	76.0	79.6	81.5	84.3	(30)
	5%	WB	77.7	77.3	76.1	72.3	68.7	66.2	65.1	65.6	68.9	71.8	73.7	76.7	(31)
		MCDB	83.7	83.5	82.3	78.5	74.8	71.6	69.8	71.2	74.8	78.3	80.2	83.0	(32)
	10%	WB	76.6	76.5	74.8	70.9	67.2	64.6	63.3	64.0	67.6	70.5	72.5	75.6	(33)
		MCDB	82.7	82.8	81.3	77.9	73.8	70.5	68.8	70.1	73.7	76.9	79.3	82.2	(34)
(35) Mean Daily Temperature Range	5% DB	MDBR	11.4	11.7	12.3	14.0	16.2	16.2	17.7	18.2	16.7	13.8	12.1	11.6	(35)
		MCDBR	12.9	13.7	13.8	15.1	16.8	16.9	18.3	19.5	19.0	15.7	14.0	13.5	(36)
	5% WB	MCWBR	4.6	4.8	5.1	5.9	7.4	7.7	8.8	9.1	8.0	6.2	5.4	5.0	(37)
		MCWBR	11.8	12.1	11.8	12.7	14.0	13.2	14.4	15.5	14.7	13.8	12.5	12.1	(38)
(40) Clear-Sky Solar Irradiance	taub		0.393	0.389	0.375	0.347	0.325	0.317	0.308	0.319	0.356	0.378	0.395	0.395	(40)
		taud	2.521	2.537	2.563	2.629	2.638	2.654	2.659	2.613	2.497	2.474	2.452	2.491	(41)
	Ebn at Noon		302	298	294	290	284	280	287	294	295	298	299	302	(42)
		Edh at Noon	36	35	32	28	25	24	24	28	34	36	38	37	(43)
(44) All-Sky Solar Radiation	RadAvg		2101	1950	1738	1527	1296	1121	1269	1542	1872	2068	2182	2183	(44)
	RadStd		148	203	146	128	72	88	97	104	140	147	206	210	(45)

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (12 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page